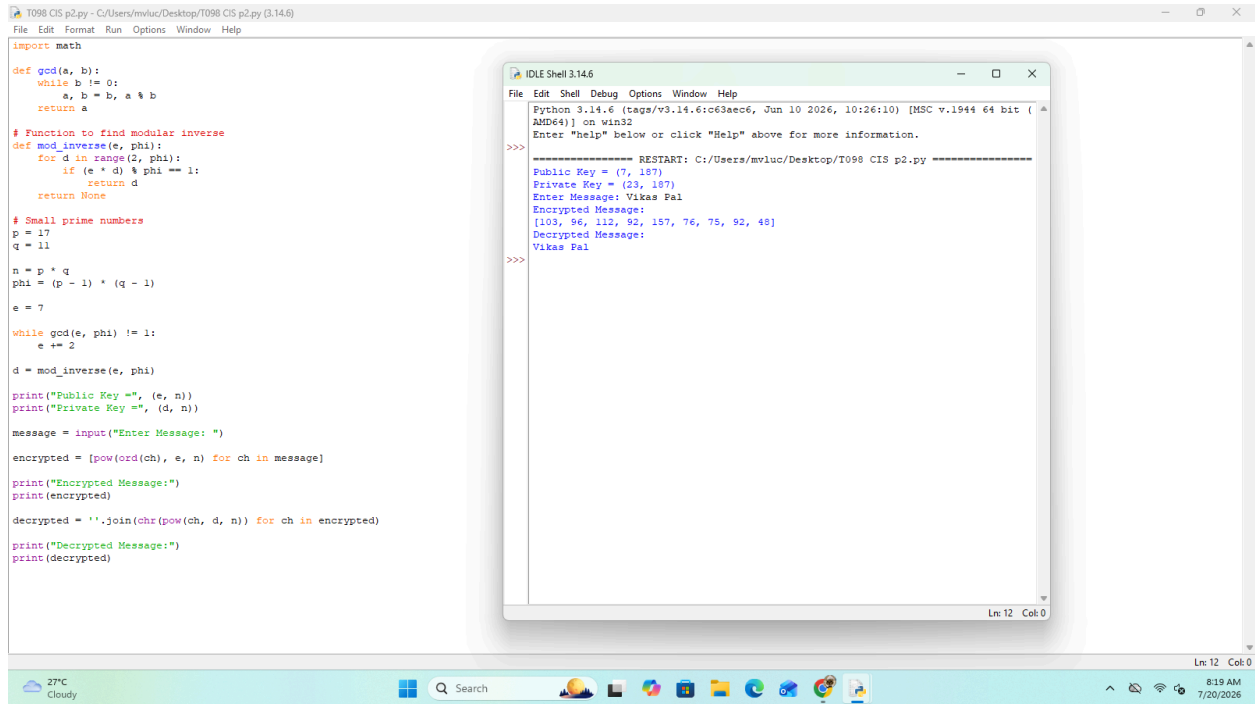


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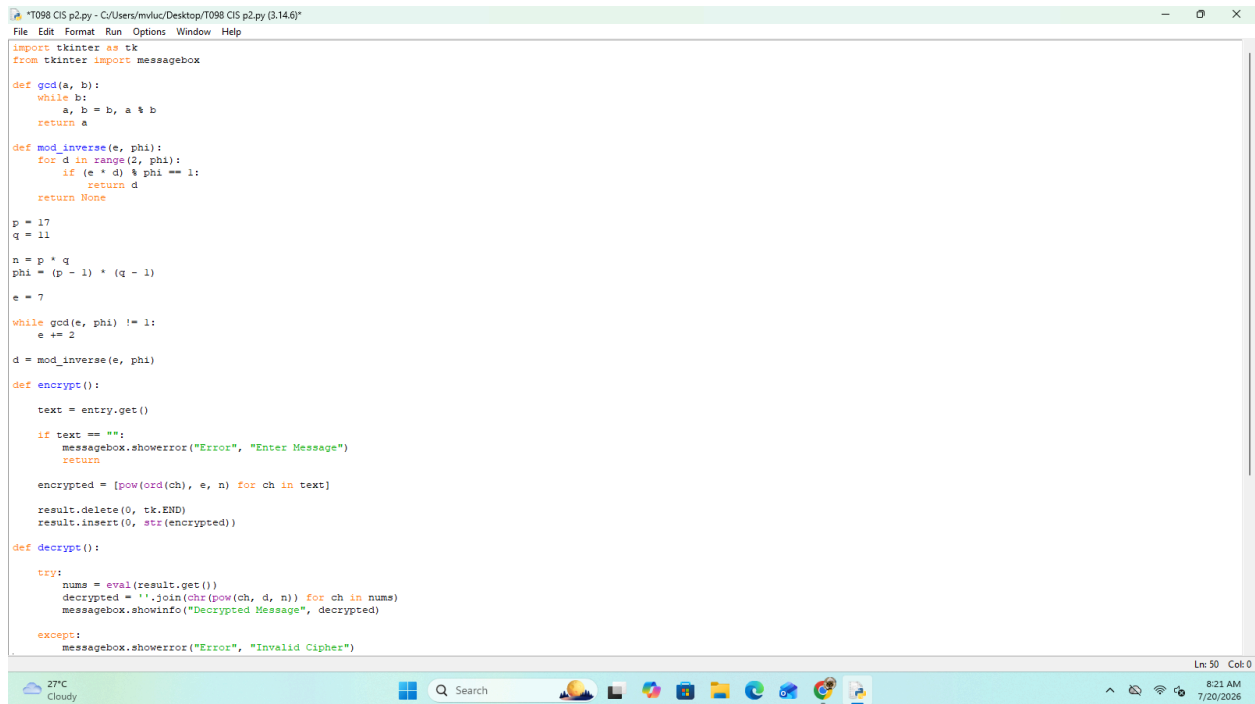
Aim:-Implement the RSA algorithm for public-key encryption and decryption, and explore its properties and security considerations.

CLI:-



The screenshot displays a Windows desktop environment. In the background, a text editor window titled 'T098 CIS p2.py - C:/Users/mvluc/Desktop/T098 CIS p2.py (3.14.6)' contains a Python script implementing the RSA algorithm. The script defines functions for calculating the Greatest Common Divisor (gcd), modular inverse (mod_inverse), and performs encryption and decryption using the formula $c = m^e \pmod{n}$ and $m = c^d \pmod{n}$. It uses small prime numbers p=17 and q=11, and a message 'Vikas Pal'. In the foreground, an 'IDLE Shell 3.14.6' window shows the execution of the script. It displays the calculated Public Key (7, 187) and Private Key (23, 187), the encrypted message [103, 96, 112, 92, 157, 76, 75, 92, 48], and the successfully decrypted message 'Vikas Pal'.

GUI:-



This screenshot shows the same Python script as the CLI version, but adapted for a graphical user interface (GUI) using the Tkinter library. The script defines functions for gcd, mod_inverse, and encryption/decryption. The encryption function uses an Entry widget for message input and a Text widget for displaying the encrypted result. The decryption function uses a Text widget for input and a Text widget for displaying the decrypted result. Both functions include error handling with messagebox.showerror. The script uses the same RSA parameters (p=17, q=11) and message ('Vikas Pal') as the CLI version. The screenshot shows the script in the text editor, but the GUI window is not visible, only the shell output from the previous run is shown in the background.

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